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ト書言

Q1

$T(n) = T(n/2) + O(1)$ . Unrolling: after  $k$  halvings the array has  $n/2^k$  elements; we stop when  $n/2^k = 1$ , i.e.  $k = \log_2 n$ . Each level does  $O(1)$  work, so  $T(n) = O(\log n)$ . Worst case: the element is absent, and we probe until the range is empty — still  $\log_2 n + 1$  probes.

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Q2

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Q3

Let  $L[i]$  = length of the LIS ending exactly at  $a[i]$ .

$L[i] = 1 + \max\{ L[j] : j < i, a[j] < a[i] \}$  (or 1 if none). Answer  $\max L[i]$ .

Two nested loops give  $O(n^2)$ . Refinement: keep  $tails[k]$  = smallest tail of an increasing subsequence of length  $k+1$ ; binary-search each element:  $O(n \log n)$ .

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Q4

True. Invariant: when BFS finishes layer  $d$ , every vertex at distance  $d$  has been discovered with  $\text{dist} = d$ . Base: layer 0 is  $\{s\}$ . Step: any  $v$  at distance  $d+1$  has a neighbour  $u$  at distance  $d$ ; when  $u$  is dequeued  $v$  is discovered with  $\text{dist } d+1$ , and no earlier layer can reach  $v$  or its distance would be  $< d+1$ .